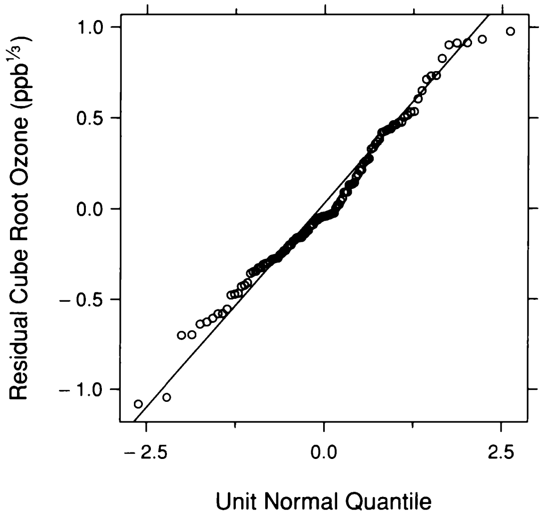
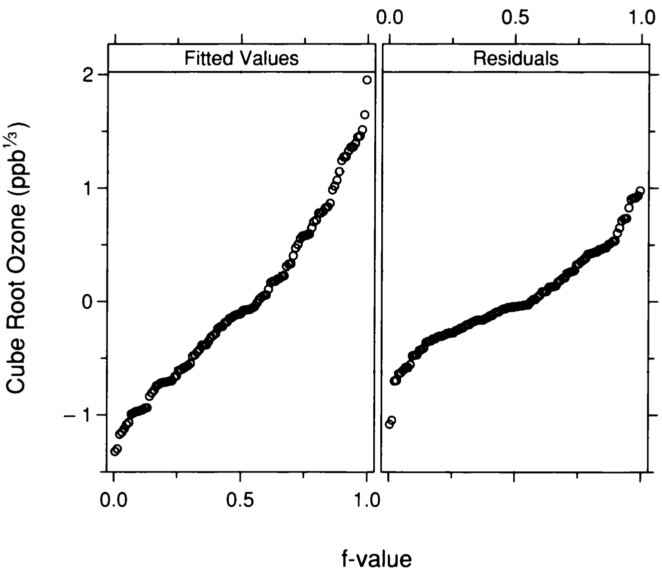
Figure 5.15 is a normal q-q plot for the fit to the cube roots; the distribution of the residuals appears to be well approximated by the normal distribution. Because of this, bisquare was not used in the loess fit. Figure 5.16 is an r-f spread plot. The loess fit has accounted for a sizeable amount of the variation in the data.




5.15 The normal q-q plot compares the normal distribution with the distribution of the residuals from the loess fit to $O_3$.



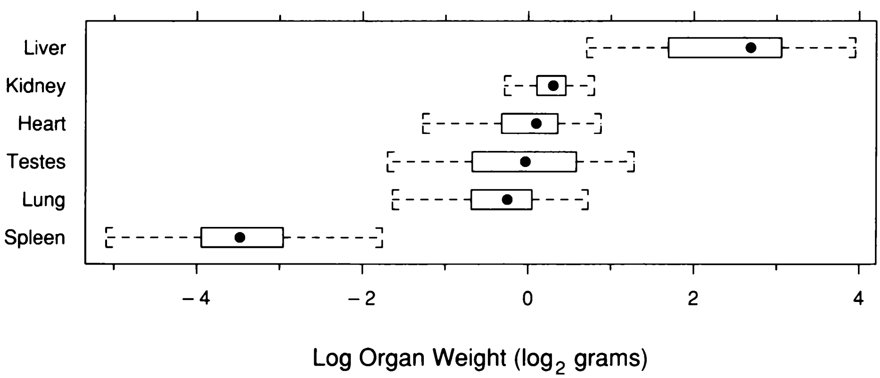

5.16 The r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the loess fit to $O_3$.

## 5.4 *Multivariate Distributions*

The environmental measurements are factor-response data; the goal is to determine how one variable, $O_3$, depends on the other three. But for many bivariate, trivariate, and hypervariate data sets, the goal is simply to determine the *multivariate distribution* of the data in the multi-dimensional space of the measurements, rather than determining how the variation in one variable depends on the variation in the others. Methods for visualizing bivariate distributions were given in Chapter 3. One example presented there is the wind speed and temperature measurements that form part of the environmental data. Another example, which we will analyze now, is data with six variables — measurements of the weights of six organs for 73 hamsters from a strain with a congenital heart problem [66]. These hypervariate data consist of 73 points in six-dimensional space.

### *Visualizing Univariate Distributions*

One aspect of any multivariate distribution is the univariate distributions of the individual variables. Figure 5.17 uses box plots to visualize the univariate distributions of the logarithms of the hamster organ weights. Logs are taken because it is more informative to consider ratios of weights. The distributions for lung, heart, kidney, and testes have similar medians, but the liver median is much larger and the spleen median is much smaller. The spreads vary, but they do not increase or decrease with the medians. For the untransformed weights, the spreads increase with the medians, so the log transformation has removed monotone spread.




5.17 Box plots display the univariate distributions of the log organ weights.

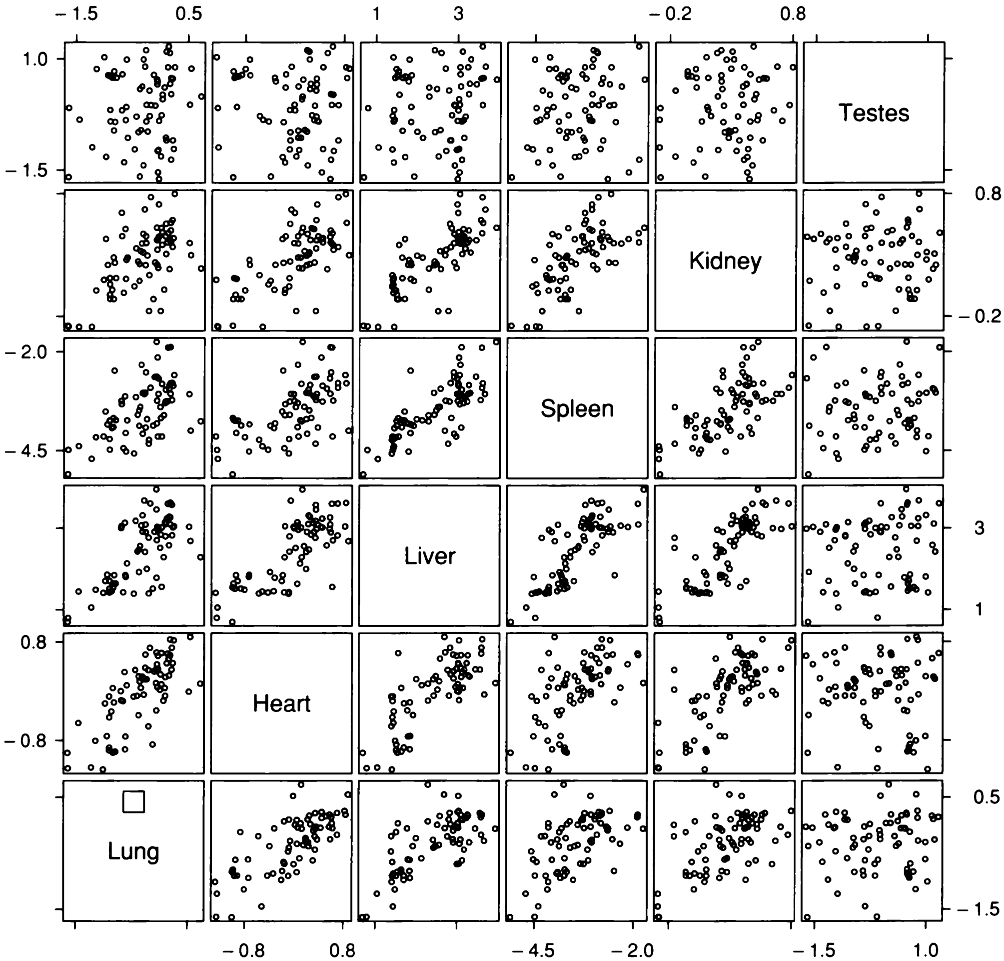
*Visualizing Bivariate and Higher-Dimensional Distributions*

Another aspect of a multivariate distribution is the bivariate distributions of all pairs of variables. Figure 5.18, a scatterplot matrix of the log organ weights, provides much information about the bivariate distributions. Lung, heart, liver, spleen, and kidney weights are correlated to varying degrees; for example, the liver and spleen weights are highly correlated. The testes weights, however, are not correlated with any of the other organ weights. That is, the testes sizes of hamsters are not related to the sizes of other body organs.

Because the scatterplot matrix allows linking by visual scanning from one panel to the next, we have the opportunity to detect aspects of multivariate distributions that involve more than just the individual bivariate distributions. But the effectiveness of linking can often be substantially increased by the enhanced linking operation of brushing, which was introduced in Chapter 4. This is the case for the log organ weights, as shown in the next section.

## 5.5 Direct Manipulation: Enhanced Linking by Brushing

The (3,4) panel of Figure 5.18, a scatterplot of log spleen weight against log liver weight, shows an outlier: the point to the northwest of the main cloud. The hamster that produced the outlier had either an enlarged spleen and a normal liver, or a small liver and a normal spleen, or perhaps even both a small liver and an enlarged spleen. The hamster has one observation on each panel of the matrix. We need to see all of these observations to help us determine if the hamster had a spleen or liver problem. Scanning to other panels from the outlying point on the (3,4) panel does not readily provide the linking because the density of the data does not let us unambiguously determine the hamster's data on the other panels. But a brush lurking in the (1,1) panel is ready to do its job.



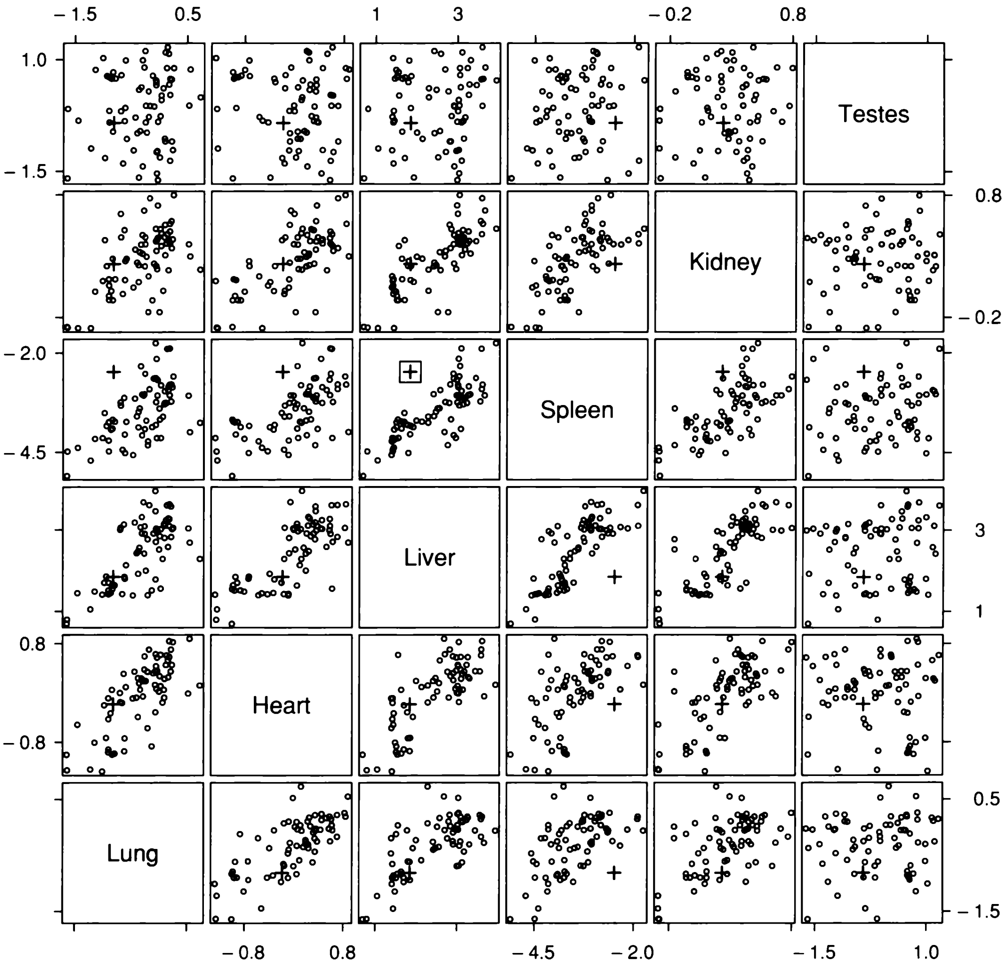

5.18 A scatterplot matrix displays the log organ weights. A brush is resting in the (1,1) panel ready to carry out enhanced linking.

In Figure 5.19, the brush is in the (3,4) panel and covers the outlier. The enhanced linking operation of brushing causes all of the data for the unusual hamster to be graphed by a "+". Row four from the bottom shows the hamster's log spleen weight is large given each of its other log organ weights; for example, its spleen is large compared with the spleens of hamsters that have about the same lung weight. But the panels of row three show that its liver weight is unusual only on the spleen plot. The hamster had an enlarged spleen.

## 5.6 *Improvisation*

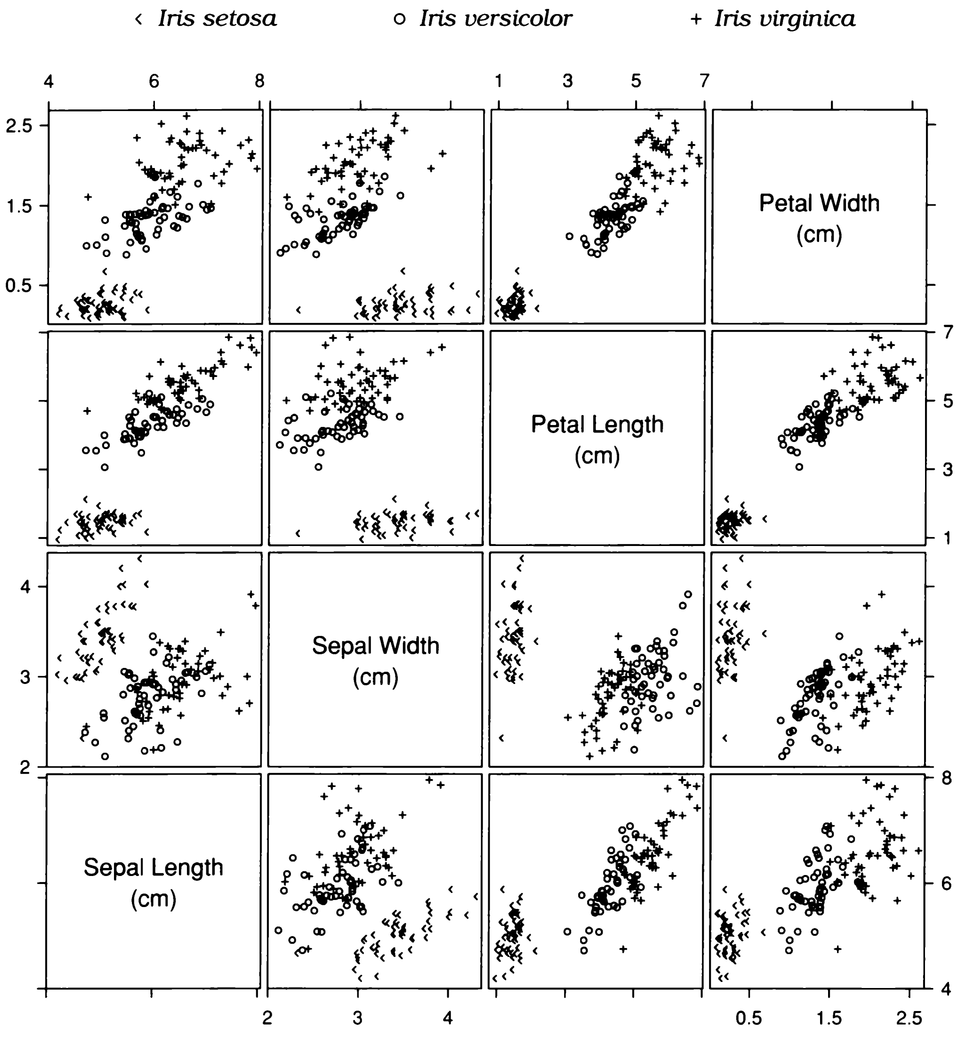
Many of the applications of visualization in this book give the impression that data analysis consists of an orderly progression of exploratory graphs, fitting, and visualization of fits and residuals. Coherence of discussion and limited space necessitate a presentation that appears to imply this. Real life is usually quite different. There are blind alleys. There are mistaken actions. There are effects missed until the very end when some visualization saves the day. And worse, there is the possibility of the nearly unmentionable: missed effects.

One important aspect of reality is improvisation; as a result of special structure in a set of data, or the finding of a visualization method, we stray from the standard methods for the data type to exploit the structure or the finding [3, 30, 76]. A few examples have been given in earlier chapters. Here we present another.




5.19 The enhanced linking operation of brushing shows all observations for one hamster.

Figure 5.20 graphs measurements of four variables — sepal length, sepal width, petal length, and petal width — for a collection of 150 irises [1]. The iris petals are the colorful parts of the flowers that we enjoy when they bloom. The sepals are the leaf-like coverings that protect the petals and other flower parts before blooming. The data, which have been jittered because of exact overlap, consist of 50 irises from each of three varieties: *Iris setosa*, *Iris versicolor*, and *Iris virginica*. Variety




5.20 A scatterplot matrix displays the iris data.

is a categorical variable and its levels are encoded by *texture symbols*, which enhance our ability to visually separate the three sets of data [21].

Despite the jittering and the use of the texture symbols, not every plotting symbol appears with clarity because so many symbols are shown, 1800 in all. The most severe problems occur in panels (3,4) and (4,3). We cannot identify every symbol in the tight cluster that appears to contain only *setosa*. But linking provides a solution. Starting at panel (3,4) and scanning to both of the panels to the left, we can see that the observations are indeed *setosa*.

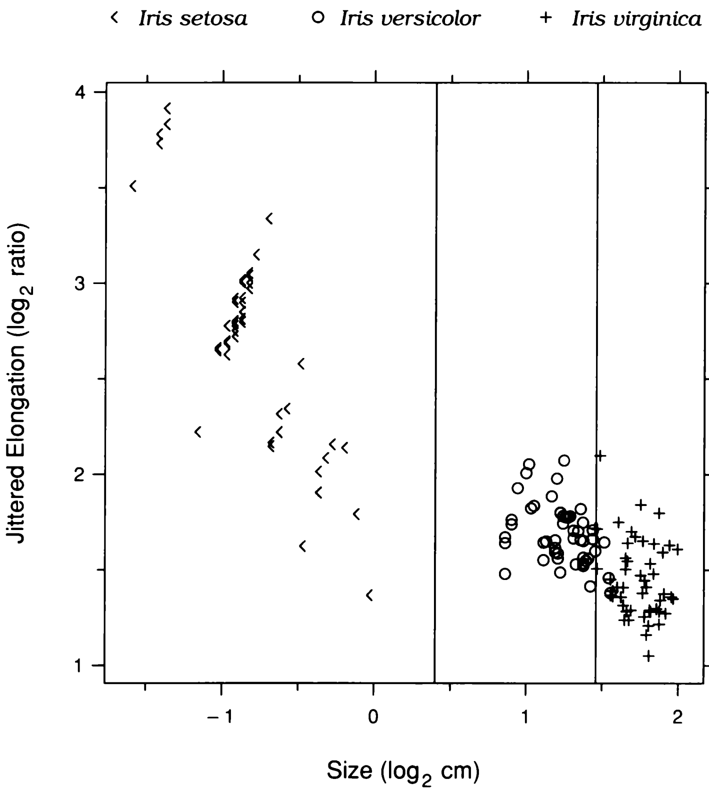
Panel (2,3) reveals an outlier: one iris in the *setosa* category with an unusually small sepal width. In this case, we can scan to find the corresponding observations on the other panels; unlike for the hamster data, we do not need the enhanced linking of brushing. First, if we scan up and down in column 2, we clearly see the observations for the unusual iris on all panels. Similarly, we can see them clearly on all panels of row 2. The positions on all other panels can be determined by scanning from both row 2 and column 2, forming cross hairs of sorts. For example, we can fix the position on panel (1,3) by scanning up from panel (1,2) and scanning to the left from panel (2,3). The scanning reveals that the sepal width is the only aspect of the iris that is unusual.

### *Classification*

The goal in analyzing the iris data is to develop a rule based on the widths and lengths of petals and sepals that would assist in determining the variety of an iris. Panel (4,3) of Figure 5.20 shows interesting behavior that suggests there might be a simple rule. The three varieties are well separated along an axis that runs from the lower left corner to the upper right. Along this axis, *setosa* is completely separated from the others, and the amount of overlap of *versicolor* and *virginica* is small. From the lower left to the upper right along this axis, both the petal length and the petal width increase, so the axis is a measure of petal size. The petals of *setosa* are smaller than those of *versicolor*, which in turn tend to be smaller than those of *virginica*.

A simple measure of petal size is the product of length and width. Also, the ratio of length to width is a measure of petal elongation. This triggers an idea, an improvisation that arises from observing the size effect. If we take the logs of both petal length and width, the mean of the two values for each flower is another size measure: the log of the square root of the above product. And the difference of the logs is another elongation measure: the log of the above ratio. It makes sense to graph one measure against the other. But this is just a Tukey m-d plot of the log width and log length.

The m-d plot is shown in Figure 5.21. The graph reveals several interesting patterns. First, the elongation tends to decrease as the size increases; in other words, as the petals become bigger they also become




5.21 A measure of petal elongation is graphed against a measure of petal size.

less elongated. Interestingly, the pattern occurs both within and between varieties. Second, size completely separates *setosa* from the other two varieties, and it nearly separates *versicolor* and *virginica*. A classification rule is shown by the two vertical lines in Figure 5.21, which are drawn at

$$c_1 = 0.4 \log_2 \text{cm}$$

and at

$$c_2 = 1.46 \log_2 \text{cm} .$$

The variety *setosa* is chosen if size is less than $c_1$; *versicolor* is chosen if size is between $c_1$ and $c_2$; and *virginica* is chosen if size is above $c_2$. These two values minimize the errors of classification for the 150 irises in the data set. Only four of the irises are misclassified by the rule, so the petal size measure provides good assistance in variety identification, at least for this set of data.

## 5.7 Visualization and Probabilistic Inference

The iris data have a long and interesting history. They were originally collected by a botanist who reported them in 1935 [1]. In 1936, R. A. Fisher used the data to illustrate his mathematical method of classification [37]. After Fisher's treatment, the iris measurements became canonical data that have been used by a multitude of people as an example to illustrate other mathematical methods for analyzing multivariate data.

The irony is that visualization of the iris data reveals more about their structure than the previous long list of analyses by numerical statistical methods. For example, the reliable classification into variety simply by petal size, and the existence of an outlier, are two insights missed by these purely numerical methods of probabilistic inference. One wonders if the iris data would have become a canonical data set had they been visualized in 1936, revealing so thoroughly to all the structure of the data.